Journal of Financial Crises

Volume 3 | Issue 3

2021

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Recommended Citation
Unnava, Vaasavi (2021) "Prompt Recapitalization Act," *Journal of Financial Crises*: Vol. 3 : Iss. 3, 254-284.
DOI: https://doi.org/10.17132/2693-3179.1228
Available at: https://elischolar.library.yale.edu/journal-of-financial-crises/vol3/iss3/15

This case study is available in Journal of Financial Crises: https://elischolar.library.yale.edu/journal-of-financial-crises/vol3/iss3/15

Prompt Recapitalization Act[1,2]

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Yale Program on Financial Stability Case Study
November 12, 2021

Abstract

In 1997, Japan's banks were in crisis due to hundreds of billions of dollars of non-performing real estate loans. In response, the government performed three rounds of capital injections in 1998, 1999, and the early 2000s. The capital injection of 1999, authorized by the Prompt Recapitalization Act, made as much as ¥25 trillion ($208 billion) available to financial institutions that applied, regardless of their capitalization. By the end of the injection window, 32 banks and trusts applied for and received ¥8.6 trillion ($71.6 billion) total in preferred shares and subordinated debts. The Act required banks to submit and adhere to restructuring plans in order to receive capital, leading to a series of mergers and acquisitions. However, differing accounting methodologies, evergreening, and double gearing allowed for systemic undercapitalization even with injections intended to help institutions meet reserve requirements.

Keywords: capital injection, double gearing, evergreening, Financial Reconstruction Commission, Japan, Japanese Financial Crisis, *jusen*, non-performing loans, Resolution and Collection Corporation, zombie lending

[1] This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering the responses to the global financial crisis that pertain to broad-based capital injection programs.

Cases are available from the *Journal of Financial Crises* at https://elischolar.library.yale.edu/journal-of-financial-crises/.

[2] The Prompt Recapitalization Act is also referred to in Japanese financial crisis literature and government documents as the Banking Recapitalization Act, Early Strengthening Act, and Financial Function Early Strengthening Act, written as **金融早期健全化法** in Japanese.

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Prompt Recapitalization Act

At a Glance

In 1997, the Japanese financial crisis began with the ballooning of non-performing loans in the financial system. After a capital injection in spring 1998 believed to be ineffective and under-allocated, the Japanese Diet passed the Prompt Recapitalization Act on October 13th, 1998.

The Act allocated ¥25 trillion ($208 billion) of capital to be injected into any bank and some non-bank financial institutions that applied. The Financial Reconstruction Commission (FRC) reviewed each application. The applications required that applicants submit management improvement plans in addition to information on capital requests. These management improvement plans factored into the FRC's prescribed underwriting terms for the capital injection—taking place in the form of preferred shares and subordinated debt—purchased by the Resolution and Collection Corporation (RCC), a subsidiary of the Depository Insurance Corporation of Japan (DICJ). The DICJ funded RCC's purchasing through a series of government-backed agency bond issuances, with leeway to borrow from the Bank of Japan as needed. There was no explicitly defined repurchase schedule.

Summary of Key Terms	
Purpose: restore both domestic and foreign confidence in Japan's financial system by disposing of non-performing loans on the balance sheets of financial institutions.	
Announcement Date	August 5, 1998
Operational Date	October 16, 1998
Injection Start Date	March 31, 1999
End of Application Window	March 31, 2001
Program Size	¥25 trillion ($208 billion)
Peak Utilization	¥8.6 trillion ($71.6 billion)
Eligibility	Any financial institution; some nonbank financial institutions
Participants	32 financial institutions
Administrator	Resolution and Collection Corporation of Japan
Legal Authority	Passed through the Japanese Diet; executed by the Prime Minister's Office and DICJ

Between March 1999 and March 2002, 32 banks applied for capital injections; no applications were rejected. Overall, of the ¥25 trillion allocated, ¥8.6 trillion ($71.6 billion) was used to purchase preferred shares and subordinated bonds. Applicant banks, trusts, and regional banks received varying capital underwriting terms dependent on the submission of management improvement plans. By mid-2015, all banks but one had repurchased all shares and subordinated debt.

Summary Evaluation

Experts view the 1999 capital injection as more successful than that of 1998. However, there are differing views on whether the system remained undercapitalized following the injection. While some scholars argue the system was fully capitalized, others note the systemic underreporting of non-performing loans on balance sheets, propagated by governmental intervention as well as the practices of zombie lending (extending capital to non-performing firms to disguise non-performing loans) and double gearing (cyclical asset purchasing to increase capital on balance sheets) prevented full capitalization of the Japanese financial sector.

Japan Context 1998–1997	
GDP (SAAR, nominal GDP in LCU converted to USD)	$4.35 trillion in 1997 $4.52 trillion in 1998
GDP per capita (SAAR, nominal GDP in LCU converted to USD)	$35,022 in 1997 $31,903 in 1998
Sovereign credit rating (5-year senior debt)	Data for Q4 1997: Fitch: AAA Moody's: Aaa S&P: AAA Data for Q4 1998: Fitch: AAA Moody's: Aa1 S&P: AAA
Size of banking system	$9.86 trillion in 1997 $9.73 trillion in 1998
Size of banking system as a percentage of GDP	226.5% in 1997 215.1% in 1998
Size of banking system as a percentage of financial system	84.1% in 1997 79.2% in 1998
5-bank concentration of banking system	42.6% in 1997 43.6% in 1998
Foreign involvement in banking system	Data not available for 1997–1998
Government ownership of banking system	Data not available for 1997–1998
Existence of deposit insurance	Yes in 1997 Yes in 1998
Sources: Bloomberg, World Bank Global Financial Development Database, World Bank Deposit Insurance Dataset.	

I. Overview

Background

The Japanese financial crisis began in November 1997, after the default of a midsized securities firm, Sanyo Securities, leading to Japan's first interbank loan default. Within weeks, Hokkaido Tokushoku, a major bank in Japan, was forced to declare bankruptcy due to inability to borrow in the interbank market, followed by the securities dealer Yamaichi Securities. Within the month, Tokuyo City Bank also failed. By December 1997, the government determined a necessity for public intervention in the financial market, but simultaneously permitted a change in accounting methodology where banks' real estate assets regularly listed on balance sheets as historical acquisition prices rather than book value, allowing banks to artificially inflate the value of the loans on their balance sheets (Hoshi and Kashyap 2010). The decision led to an eventual loss of confidence in the accounting and auditing system in Japan, as the true amount of bad loans at failed financial institutions exceeded the amount published prior to the failure of these institutions (Fukao 2000).

While the Japanese financial system faced difficulties in the early 1990s, there was no formalized capital injection framework, as Japan had never suffered a banking crisis in the post-war era (Nakaso 2001). The Japanese Diet, the legislative body of Japan, passed its first measure of intervention to counteract the banking crisis in February 1998 with the Financial Function Stabilization Act. The act allocated ¥30 trillion of public funds to support the banking sector, of which ¥13 trillion was specifically allocated for bank recapitalization (Hoshi and Kashyap 2010).

Experts viewed the capital injection of March 1998 as unsuccessful, with only ¥1.8 trillion utilized out of the ¥13 trillion allocated, spread across 21 institutions (Hoshi and Kashyap 2010). Banks were reluctant to participate in the program, unwilling to be singled out as a weaker bank. Ultimately, all major banks applied for capital injections simultaneously to disguise any signaling of weakness (Nakaso 2001). The healthiest bank of the 21 institutions applied for only ¥100 billion in capital with only one bank applying for more. As a result, few if any banks received enough funding to become well-capitalized (Hoshi and Kashyap 2010). In addition, there was no thorough clean-up of bank balance sheets for those banks seeking public capital injections (Fukao 2000). Banks originally shunned the program, but government pressure finally resulted in each of the major banks applying for the same amount of public funds. There were two possible reasons for banks refusing public funds. First, banks may have feared applying for public funds in sending negative signals about their respective balance sheets pushing down the value of existing equity. Banks may also have feared the seniority of new securities purchased by the government to existing security claims (Hoshi and Kashyap 2010).

Public dissatisfaction over the handling of the financial crisis mounted, leading to a show of public distrust in the current Diet with the election of the Liberal Democratic Party to majorities in both houses of the Diet as well as the Prime Ministership (Hoshi and Kashyap

2010). In summer 1998, issues in the financial system proved persistent with the stock price of Long-Term Credit Bank falling sharply after a rejected merger between itself and Sumitomo Trust and Banking (Fukao 2000).

Many major Japanese banks still were under-capitalized following the execution of the Financial Function Stabilization Act. In response, the plenary session of the Japanese Diet began with the Liberal Democratic Party announcing a forthcoming package of bills aimed to combat the unstable Japanese financial landscape. The Prime Minister's Office on August 5, 1998, introduced a package of five financial stabilization bills to the Diet (*Japan Times* 1998a). On October 16, 1998, the Japanese Diet passed the Prompt Recapitalization Act, the final form of the bank recapitalization bill (*Japan Times* 1998e). A timeline of legislation passed during 1998 is in Figure 1 below.

Figure 1: Timeline of Legislation

February 16, 1998	Financial Function Strengthening Act
June 5, 1998	Financial Reconstruction Law
Mid-Early October 1998	Establishment of FRC Revolving Credit Transfers Facilitation Act Financial Services Agency Establishment Act Amendment to Deposit Insurance Act Measures regarding debt management collection business
Mid-Late October 1998	Prompt Recapitalization Act
December 1998	FRC Inaugurated

Source: Nakaso 2001.

Program Description

The Prompt Recapitalization Act was intended to restore both domestic and foreign confidence in Japan's financial system by disposing of non-performing loans on the balance sheets of financial institutions (Prompt Recapitalization Act 1998). The Act also increased the budget for capital injections into undercapitalized banks from ¥13 trillion under the previous law to ¥25 trillion ($208 billion) (Hoshi and Kashyap 2010). [4]

The recapitalization was overseen by the Financial Reconstruction Commission, created by the Financial Reconstruction Commission Establishment Law to oversee operations under the suite of new laws passed for financial stabilization in October 1998 (Nakaso 2001; Prompt Recapitalization Act 1998). The FRC was an independent commission and arm to the Prime Minister's office called the Financial Reconstruction Commission (FRC). The FRC was independent of the Bank of Japan or the Treasury. Five members were appointed to this committee, with the chairman appointed from the prime minister's cabinet (Nakaso 2001). The FRC was required to terminate its job of special administration on March 31, 2001 (Iimura 1999).

The Financial Revitalization Act[5], a sister act passed in the same month as the Prompt Recapitalization Act, established the Resolution and Collection Committee (RCC)[6], an asset management corporation, as a subsidiary of the Depository Insurance Corporation of Japan (DICJ) (DICJ 2021b). Funded by the DICJ, the RCC purchased either preferred stocks or subordinated bonds from the financial institutions that applied for capital injections under the new scheme (Hoshi and Kashyap 2010; DICJ n.d.a).

No banks were excluded from eligibility under the Prompt Recapitalization Act. Three non-bank financial institutions, all cooperatives, were also eligible, referred to by name in the law: Norinchukin Bank; the Agricultural Cooperative Association; and the Federation of Fisheries Cooperative Associations. There were also no barriers to foreign banks participating, although these foreign banks were subject to more stringent capital adequacy requirements in comparison to their domestic counterparts (Prompt Recapitalization Act 1998). Each organization faced separate capital adequacy ratio definitions dependent on their status as bank holding companies, financial institutions, or financial institutions with subsidiaries as well as their status as domestic or foreign institutions.

In addition to no exclusions from eligibility for this intervention, there were no explicit capital limits written for participating banks.

[4] In addition to the ¥25 trillion allocated for recapitalization, the government allocated an additional ¥18 trillion for the nationalization of failed banks under the Financial Revitalization Act (Fukao 2000, 4). Combined with ¥17 trillion in depositor protection, this totaled to a ¥60 trillion package for financial stabilization (Hoshi and Kashyap 2010, 409).

[5] The Financial Revitalization Act is also referred to as the Act for Rehabilitation for Financial Functions. It was Act No. 132 during Heisei 10, or 1998, written as 金融機能の再生のための緊急措置に関する法律 in Japanese.

[6] For more information regarding the RCC and its operations, please refer to the "The Resolution and Collection Corporation of Japan" authored by Mallory Dreyer (2021).

Banks requesting capital funding applied through the newly formed FRC and had to provide restructuring plans to demonstrate how they planned to improve performance. These plans were made available to the public (Nakaso 2001). In addition to restructuring personnel, banks were asked to reduce the number of foreign branches (Fukao 2000). Banks with higher capital adequacy ratios that sought funding were required to absorb failing banks through mergers and acquisitions (*Japan Times* 1998d).

These additional constraints partially determined the terms for preferred shares purchased by the RCC through capital injection. After receiving the request for a capital injection, the FRC would assess the performance of the bank, nature of the instrument for injection, and the management improvement plan to determine the appropriate cost of capital for each bank (Nakaso 2001). After these terms were decided by the FRC, the RCC would purchase and ultimately manage the shares and subordinated debt using capital from the DICJ Early Strengthening Account, a fund set up for injections under the Prompt Recapitalization Act, also called Act for the Early Strengthening of Financial Functions (DICJ 2021b; n.d.a).

Outcomes

In the initial phase of capital injections, 15 of the 16 major banks in Japan participated, requesting a total of ¥4.8 to ¥5.7 trillion (Naito 1999). After evaluation, the 15 banks were rewarded more capital than requested, totaling ¥7.46 trillion based on assessed need (Fukao 2000; Nakaso 2001).[7] Over the next two years, an additional 17 institutions applied for capital injections, bringing the total to ¥8.6 trillion: ¥7.32 trillion in preferred shares and ¥1.32 trillion in subordinated bonds (DICJ n.d.a). The FRC did not turn down any applications (Hoshi and Kashyap 2010). This injection exceeded the capital injection of 1998, for which 21 institutions applied for a total of just ¥1.8 trillion.

Almost every bank repurchased their shares and debt sold to the DICJ under this program, as shown in Figures 1 and 2 below. Shinsei Bank remains the exception; formerly the Long-Term Credit Bank nationalized under the Financial Reconstruction Act, Shinsei Bank continues to list the Resolution and Collection Commission as a major shareholder. As of March 31, 2020, the DICJ and RCC hold 18.1% of Shinsei bank's common shares (Shinsei Bank 2021).

[7] For more information on the terms of lending and private capital fundraising information, please refer to Tables 1 and 2 in (Fukao 2000).

Figure 2: Timeline of preferred share repurchasing

Bank	Injection Date	Final Repurchase Date
Dai-ichi Kangyo Bank	March 1999	July 2006
Fuji Bank	March 1999	July 2006
Industrial Bank of Japan	March 1999	August 2005
Sakura Bank	March 1999	October 2006
Sumitomo Bank	March 1999	September 2006
Sanwa Bank	March 1999	May 2006
Tokai Bank	March 1999	June 2006
Toyo Trust & Banking	March 1999	June 2006
Mitsubishi Trust & Banking	March 1999	January 2001
Daiwa Bank	March 1999	March 2009
Asahi Bank	March 1999	June 2015
Sumitomo Trust & Banking	March 1999	March 2013
Chuo Trust & Banking	March 1999	March 2013
Bank of Yokohama	March 1999	August 2004
Ashikaga Bank	September & November 1999	February 2006
Hokuriku Bank	September 1999	July 2009
Hokkaido Bank	March 2000	August 2009
Bank of the Ryukyus	September 1999	July 2010
Hiroshima-Sogo Bank	September 1999	December 2005
Kumamoto Family Bank	February 2000	May 2006
Shinsei Bank*	March 2000	—
Chiba Kogyo Bank	September 2000	July 2013
Yachiyo Bank	September 2000	March 2006
The Nippon Credit Bank	October 2000	June 2015
Kansai Sawayaka Bank	March 2001	October 2003
Higashi-Nippon Bank	March 2001	March 2011
Kinki Osaka Bank	April 2001	June 2015
Gifu Bank	April 2001	December 2010
Fukuoka City Bank	January 2002	July 2010
Wakayama Bank	January 2002	December 2005
Kyushu Bank	March 2002	February 2008

* Shinsei Bank is the successor of the nationalized Long-Term Credit Bank, nationalized

under the Financial Reconstruction Act.

Source: Deposit Insurance Corporation of Japan 2020b.

Figure 3: Timeline of subordinated debt repurchasing.

Bank	Injection Date	Final Repurchase Date
Dai-ichi Kangyo Bank	March 1999	March 2005
Fuji Bank	March 1999	September 2004
Industrial Bank of Japan	March 1999	March 2004
Sanwa Bank	March 1999	September 2005
Asahi Bank	March 1999	March 2009
Sumitomo Trust & Banking	March 1999	January 2004
Mitsui Trust & Banking	March 1999	March 2005
Bank of Yokohama	March 1999	May 2004
Kansai Sawayaka Bank	March 2001	January 2004

Source: Deposit Insurance Corporation of Japan 2020b.

II. Key Design Decisions

1. The Prompt Recapitalization Act was introduced in conjunction with other financial revitalization legislation, though not explicitly part of a package.

The Japanese Diet passed a series of financial revitalization bills on October 2, 1998 (*Japan Times* 1998c). Among these bills were the Financial Reconstruction Committee Establishment Law, the Revolving Credit Transfers Facilitation Act, the Financial Services Agency Establishment Act, an amendment to the Deposit Insurance Act, and measures regarding debt management collection businesses. The evening of the passage of this set of bills, the Liberal Democratic Party proposed a plan for bank recapitalization to be debated by the upper and lower houses of the Japanese diet (*Japan Times* 1998b). The bill was passed 14 days later on October 16, 1998 (*Japan Times* 1998e).

The legislation was part of a series of capital injections between 1998-2008, preceded by a capital injection in March 1998, and followed by a third injection legislation in June 2004 (Hoshi and Kashyap 2010).

2. The Prompt Recapitalization Act passed formally through the Japanese Diet.

The Japanese Diet first addressed the banking crisis in February 1998 with the Financial Function Stabilization Act. This act allocated ¥30 trillion of public funds for a set of measures, including ¥13 trillion for bank recapitalization (Hoshi and Kashyap 2010). However, just ¥1.8 trillion of funds were allocated in March 1998 (DICJ 2020a).

The plenary session of the 1998-99 Japanese Diet began with the Liberal Democratic Party announcing a forthcoming package of bills aimed at the unstable Japanese financial landscape. Beginning as two financial stabilization bills introduced to the Diet by the Prime Minister's Office on August 5, 1998, the proposed stabilization program eventually evolved into the Financial Reconstruction Act—which provided the framework with which to nationalize the Long-Term Credit Bank of Japan—in addition to an act establishing the Financial Reconstruction Commission, both of which were passed through the lower house on October 2, 1998 (Iimura 1999; *Japan Times* 1998a; *Japan Times* 1998b). That evening, the Liberal Democratic Party proposed a bill for bank recapitalization (Iimura 1999). On October 13, 1998, the Japanese Diet passed the Prompt Recapitalization Act, the final form of the bank recapitalization bill (*Japan Times* 1998d).

Through the new bills, the government doubled available funds for financial revitalization from ¥30 trillion to ¥60 trillion. Of this ¥60 trillion, ¥25 trillion was available for capital injections (Nakaso 2001). These injections were done in installments as banks applied for funding (DICJ 2021a). In addition, ¥18 trillion was allocated to nationalize failed banks and ¥17 trillion was allocated for depositor protection (Hoshi and Kashyap 2010).

3. The capital injection was overseen by the FRC, an independent commission under the Prime Minister's office, but the RCC, an asset management company, purchased and managed preferred shares and subordinated debts.

Within two months of the passage of the Prompt Recapitalization Act, the Financial Reconstruction Commission (FRC) began to work on determining the dividend policies for the preferred shares to be bought for capital injections (*Japan Times* 1998f). The FRC was an independent commission under the Prime Minister's office. It had five members, including a cabinet member who served as the chairman, and oversaw the Financial Supervisory Agency. Through legislation, the Diet gave the FRC planning authority on matters concerning the resolution of financial institution failures and financial crisis management, as well as the authority to inspect and supervise financial institutions (Nakaso 1999).

By legislation, the FRC's term only lasted for two years and officially ended in January 2001 (Spindle and Dvorak 2000). At that time, the FRC and the Financial Supervisory Agency were merged to create the Financial Services Agency (FSA), now Japan's bank, insurance, and securities regulator. The FSA then took on the responsibilities of the FRC, and screened applications for the final three banks to apply for public injection through the Prompt Recapitalization Act.

The Resolution and Collection Corporation (RCC) was created as a merger between the Housing Loan Administration Corporation and the Resolution and Collection Bank on April 1, 1999, under the Financial Revitalization Act (RCC 2018; DICJ 2021b). The RCC was funded entirely by the DICJ (RCC 2018).

The RCC was a subsidiary of the DICJ (DICJ 2021b). The DICJ acts independently of the Bank of Japan or the Treasury, though in close cooperation. Financial assistance from the DICJ is funded through the issuance of government-backed DICJ bonds. In rare instances, the DICJ may borrow money directly from the Bank of Japan (FSB 2016).

The RCC used DICJ capital to purchase preferred shares and subordinated debt of banks that requested capital injections (DICJ 2005) . The RCC noted that it was able to exercise its rights as a shareholder and investor, although available information doesn't clarify to what extent the RCC exercised those rights (RCC 2018). The preferred shares converted to common shares after a grace period; if the FRC (or FSA after the merger of the two organizations) was dissatisfied by progress in restructuring for a specific bank, it could convert the shares to common shares and use its position as largest shareholder to put pressure on management (Kanaya and Woo 2000). The FRC determined the terms of the capital on a case-by-case basis (Nakaso 2001).

Enforcement in defining the underwriting terms of the preferred shares and subordinated loans faced issues due to varying leadership styles over the year and a half of the execution of the law. While the first chair of the FRC, Hakuo Yanagisawa, attempted to enforce strict terms such as calling for major restructuring of banks in exchange for capital injections, he was removed from the position in less than a year to be followed by "less-resolute" chairmen due to attempts to balance coalitions built in the Diet by Prime Minister Obuchi. The changes in leadership to the FRC caused volatility in the Japanese market, with shareholders unsure of future underwriting terms. Under the new chair, Michio Ochi, the restructuring slowed. Where Yanagisawa inserted requirements that banks meet Basel capital standards, Ochi dropped such requirements; similarly, where Yanagisawa requested banks merge, Ochi was reluctant to force mergers (Spindle and Dvorak 2000). Following a scandal after recordings surfaced with Ochi inviting bankers to approach him for more lenient standards, Ochi was followed by three other Chairmen over the one and a half years of the FRC's existence before the first chairman of the FRC, Hakuo Yanagisawa, was reappointed.

4. The recapitalization bill and requirements for recapitalization were announced publicly and debated thoroughly before the execution of the capital injection.

The bill, announced as part of a series of bills to address the *jusen* problem, was originally introduced as an idea by the Liberal Democratic Party in the plenary session of the Japanese Diet (*Japan Times* 1998a). After the passage of other financial revitalization bills, the Prompt Recapitalization Act faced much negotiation before its passage 14 days later (*Japan Times* 1998c). Components of the original proposal changed—for instance, the initial proposal contained a constraint that banks with a capital adequacy ratio of 8% or higher would only receive capital injections when there is an "imminent danger or deflationary spiral" (*Japan Times* 1998a).

5. There were no constraints on financial institutions for eligibility, though domestic banks were subject to a lower capital adequacy ratio requirement.

Any domestic or foreign bank was eligible for a capital injection (Prompt Recapitalization Act 1998). However, no foreign banks participated (DICJ 2020b).

The law did not require participation of any banks (Prompt Recapitalization Act 1998). The law also permitted applications by three specific non-banks, all cooperatives: Norinchukin Bank, Agricultural Cooperative Association, and the Federation of Fisheries Cooperative Associations.

Banks considered "capitalized" were required to acquire or merge with a struggling bank in order to receive capital injections (*Japan Times* 1998d). The definition of under-capitalization varied depending on whether the bank was a domestic or foreign entity, as shown in Figures 2, 3, and 4 below (Prompt Recapitalization Act 1998).

Figure 4. Capital Ratios for financial institutions, bank holding companies, and their subsidiaries

	Banks or credit unions with overseas sales offices, and Norinchukin Bank	**Other financial Institutions**
Classification of sound capital status	Capital adequacy ratio of 8%	Capital adequacy ratio of 4% or more according to domestic standards
Classification of under-capitalized status	Capital adequacy ratio of 4% to 8%	Capital adequacy ratio based on domestic standards of 2%-4%
Classification indicating that there is a significant under-capitalized situation	Capital adequacy ratio of 2% to 4%	Capital adequacy ratio of 1%- 2%
Classification indicating that there is particularly low capital	Capital adequacy ratio related to international unified standards 0% to 2%	Capital adequacy ratio of 0%- 1%

Source: Prompt Recapitalization Act 1998.

6. The underwriting terms of capital injections were dependent on both capitalization status and management improvement plans.

In addition to balance sheet information, banks were required to submit management improvement plans that explained structural changes they planned to make. These affected the underwriting terms for the preferred shares and subordinated bonds purchased in the capital injections. The FRC sought to encourage restructuring, cost reduction, and corporate reorganization, and including these factors in management improvement plans decreased the overall cost of government capital through reduced dividends and bond yields (Nakaso 2001). Capital was injected both through purchases of preferred shares and subordinated debt (Hoshi and Kashyap 2010). (A detailed table listing of the dividend terms for preferred shares for the first 15 banks to receive injections is available in the source note; a listing of underwriting terms for subordinated debt of the first 15 banks to receive injections is available in the source note as well.)

Preferred shares had mandatory conversion dates to common shares; however, many banks chose to purchase these preferred shares and reissue common shares before their respective mandatory conversion dates (DICJ 2020b).

In addition, banks were encouraged to close foreign branches and subsidiaries, and incorporated such stipulations into their restructuring plans. They were rewarded in underwriting requirements guided by a rubric shown below in Figure 8. The proposed restructuring plans also involved reductions of personnel (Fukao 2000). An example of these proposed reductions are in Figures 5, 6, and 7 below:

Figure 5. Proposed closing of foreign branches

Bank	Type
Bank of Yokohama	Complete withdrawal by March 1999
Daiwa Bank	Complete withdrawal by March 2000
Mitsui Trust and Banking	Complete withdrawal by March 2000
Chuo Trust and Banking	Complete withdrawal by March 2000
Toyo Trust and Banking	Complete withdrawal by March 2001

Source: Fukao 2000.

Figure 6: Proposed changes in number of foreign branches and subsidiaries

	March 1998	March 2003	Change in Number	Rate of Change (%)
City banks:				
Industrial Bank of Japan	38	28	-10	-26
Dai-Ichi Kangyo	46	31	-15	-33
Sakura Bank	46	32	-14	-30
Fuji Bank	43	27	-16	-37
Sumitomo Bank	64	36	-28	-44
Sanwa Bank	45	33	-12	-27
Tokai Bank	46	21	-25	-54
Asahi Bank	21	6	-15	-71
Mitsubishi Trust and Banking	19	10	-9	-47
Sumitomo Trust and Banking	16	6	-10	-66

Source: Fukao 2000.

Figure 7: Proposed changes in personnel and related expenditures

***millions of yen, %*

Bank	Number of Employees			Personnel Costs			Other Costs		
	March 1999	March 2003	Rate of Change	March 1999	March 2003	Rate of Change	March 1999	March 2003	Rate of Change
Industrial Bank of Japan	4776	4482	-6	68600	68000	-1	60700	49800	-18
Dai-Ichi Kangyo Bank	16130	13200	-18	165600	138300	-17	166200	149300	-10
Sakura Bank	16700	13200	-21	179900	152100	-16	195300	185700	-5
Fuji Bank	14250	13000	-9	153000	137500	-10	137000	132500	-3
Sumitomo Bank	15000	13000	-13	156100	147300	-6	137800	128900	-7
Daiwa Bank	7640	6300	-18	63000	52300	-17	91778	89569	-2
Sanwa Bank	13600	11400	-16	148400	125600	-15	144400	140900	-2
Tokai Bank	11125	9731	-13	111600	92700	-17	89705	82996	-8
Asahi Bank	12800	11800	-8	113700	107000	-6	94000	93000	-1
Bank of Yokohama	5718	4512	-21	50500	43000	-15	41700	40000	-4
Mitsubishi Trust and Banking	4932	4695	-5	68293	62640	-8	60086	59828	0
Sumitomo Trust and Banking	5900	5200	-12	61000	52000	-15	56500	53600	-5
Toyo Trust and Banking	4100	3400	-17	42300	38100	-10	30700	30000	-2
Mitsui Trust and Banking + Chuo Trust and Banking	9980	8900	-11	91600	82100	-10	78300	71600	-9
Total	142651	122820	-14	1473593	1298640	-12	1384169	1307693	-6

Source: Fukao 2000.

The management improvement plans were graded on a rubric, where points were allocated for restructuring behaviors consistent with the FRC's expectations to successfully eliminate the non-performing loan problem. An approximate translation of the rubric is in Figure 9 below:

Figure 8: Rubric for management improvement plans

Evaluation items for improvement of management soundness plan
Gain Points
1. Response to reorganization (1) Are you prepared for financial restructuring such as mergers, subsidiaries, and capital alliances? (2) Are alliances with other types of businesses conducted? 2. Business reconstruction (1) Has the regional bank completely withdrawn overseas or has it reduced overseas branches/local subsidiaries? (2) Are there specific and clear strategies for improving profitability? (3) Is the organization planning to fundamentally reform itself? 3. Restructuring (1) Has the total labor cost been reduced? (2) Has the number of officers and the number of employees been reduced ? (3) Have property costs (excluding mechanization costs) been reduced? 4. Other (1) Is the total amount of loans (excludes impact loans and actual basis) increased? (2) Is self-procurement planned? (3) Is the public fund application amount particularly sufficient? (4) Is the internal corporate rating accurate? (5) Is the liquidation of non-performing loans specifically planned? (6) Is the usage of consultants and advisors abolished?

(7) Has the average monthly salary decreased sufficiently (whether salary system has been revised)?
(8) Are dividends reduced?
Lose Points
(1) Is processing of unrealized loss on securities slow?
(2) Is there a description of the background of the occurrence of bad debts?
(3) Is the number of officers changing or increasing?
(4) Is the payment of director bonuses, remuneration, and retirement bonuses excessive?
(5) Is there an increase in property expenses?
(6) Is the disposal of idle facilities insufficient?

Source: Financial Reconstruction Commission 2001.

7. There were constraints on management pay as well as shareholder compensation.

During the course of the capital injection, management positions were no longer allowed to take bonuses on their salary. In addition, the banks were no longer allowed to issue dividends to shareholders while using capital injection funding (Prompt Recapitalization Act 1998).

8. There was no explicit exit strategy outlined or mandated for banks participating in the capital injection.

All banks receiving capital injections in the form of subordinated debt received step-up clauses that began after three years of receiving the injection. Mandatory conversion dates of preferred shares for each bank varied widely, from within three months of injection in the case of Daiwa Bank to after six years in the case of Sumitomo Bank (DICJ 2020b).

There were no explicit legislated or enforced exit strategies. However, all banks but one—Shinsei Bank, formerly the Long-Term Credit Bank of Japan—repurchased their shares by June 2015 (DICJ 2020b). The Shinsei Bank continues to list the Resolution and Collection Commission as a majority shareholder. As of March 31, 2020, the DICJ and RCC hold 18.1% of Shinsei bank's common shares (Shinsei Bank 2021). These shares resulted from a negotiation between Ripplewood Holdings and the Japanese government, as a condition to buy the Long-Term Credit Bank in 1999. While Ripplewood could not immediately offload the bad loans, they were provided an injection through the capital injection scheme, in addition to a guarantee that the Japanese government double the Long-Term Credit Bank's loan-loss reserves (Bremner 1999).

III. Evaluation

Unlike the capital injection in March 1998, the Prompt Recapitalization Act provided more capital to relieve stress due to non-performing loans. However, there is some disagreement whether the capital injection was large enough to meet estimated potential losses in the Japanese banking system. Nakaso argues that the banking system suffered from nearly ¥11.7 trillion potential losses due to non-performing loans in 1999. With the original ¥7.2 trillion of injected capital, in addition to the ¥2.1 trillion of private fundraising and the ¥2.5 trillion net operating profit, the capital injection was sufficient in capitalizing the banking sector (Nakaso 2001). However, some economists and analysts believed that underreporting of non-performing loans still occurred into 2002, after the capital injection. By 2002, the amount of non-performing loans in the banking system actually increased from the original ¥29.6 trillion in March 1999 to ¥42.0 trillion (Hoshi and Kashyap 2010). Kashyap estimates that the combined effect of the banking problems was approximately ¥40 trillion, though estimated varied broadly as seen below in Figure 10 (Kashyap 2002).

Figure 9: Experts' estimates of the insolvency of the Japanese banking system

Analyst	Firm	Estimate	Comments
David Atkinson	Goldman Sachs (October 31, 2001)	¥70 trillion of net loan losses based on March 2001 loans (¥18.7 trillion for the major banks)	Large bank losses represented 161% of capital adjusted for tax loss carry forwards and public money.
Robert Feldman	Morgan Stanley (August 2002)	¥22 trillion	Intended to be a lower bound for additional taxpayer exposure.
James Fiorillo	ING Securities (Japan) (August 2002)	¥19.9 trillion in net loan losses, -¥2 trillion in unrealized capital gains	Capital (as reported without adjustments) ¥16.2 trillion.
Yukiko Ohara	Credit Suisse First Boston Securities (Japan) Limited (July 2002)	¥21.8 trillion in required credit costs for the major banks	Estimated non-performing loans for the major banks: ¥121.9 trillion.
Paul Sheard	Lehman Brothers (August 2002)	"To restore the balance sheet health and credibility of the banking system would probably require ¥30 to ¥50 trillion."	Note that the deposit insurance fund has ¥49 trillion of untapped capacity. Thus, infrastructure and budgeting are in place to act if there was political will.
Reiko Toritani	Fitch Ratings (August 2002)	¥23 trillion for the major banks	Adjusting the stated value of equity for the major banks as of March 2002 to account for fictitious tax credits, public funds, and unrealized gains implies a market value of essentially zero.

Source: Kashyap 2002.

Despite dispute over the appropriate size of the injections, it is widely believed that the injection helped recapitalize the Japanese financial system. Nakashima and Souma (2011) find the two injections of 1998 and 1999 significantly reduced financial risks faced by banks through reduced default risks. The capital injection of 1999 succeeded in comparison to the injections of 1998 due to the risk-based liability evaluations for capital injections in the Prompt Recapitalization Act, where regulators had access to bank balance sheets, in comparison to the Financial Function Stabilization Act, where the commission overseeing injections were unable to look at bank balance sheets (Nakaso 2001; Allen, Chakraborty, and Watanabe 2009). Banks receiving capital injections in 1999 increased lending, while the 1998 capital injection had no such effects (Allen, Chakraborty, and Watanabe 2009).

In addition to the capital injections, banks merger activity accelerated. In January 1999, Chuo Trust & Banking Co. and Mitsui Trust & Banking Co. merged, cutting costs through lowering salaries in a new, unified pay scale. In the next year, 17 banks announced mergers (Spindle and Dvorak 2000).

However, some constraints in the Japanese financial system interfered with the Japanese government's ability to fully recapitalize the financial sector while successfully combating the full scale of the non-performing loan problem, leading to disputes over the true capital needs of the financial system.

First, the implementation of adjusted Basel capital standards for banks lead to systemic overstating of capitalization on bank balance sheets due to Japanese accounting practices. Japanese banks engage in long-term relationships with the businesses they fund; as banks acquired shares in these firms, unrealized gains due to increase in share prices were "hidden" from bank balance sheets by Japanese accounting standards. During negotiations determining capital standards with the Basel committee, Japanese regulators successfully argued that unrealized gains from these *keiretsu* shares should be counted towards tier-II capital standards. However, as share prices fell, actual rates of capitalization fell as well, reducing the size of Japanese banks' tier-II capital holdings (Ito and Sasaki 2002). In addition, Japanese banks tend to reserve only against recognized bad loans. In the early 2000s, the banks only capitalized enough to cover between 40 to 60 percent of all bad loans in the banking system (Hoshi and Kashyap 2004).

Additionally, the fear of falling below capital standards encouraged bank credit lending to insolvent borrowers under the belief that non-performing loans would become performing loans, or the government would bail them out (Caballero, Hoshi, and Kashyap 2008). While banks may lend capital to temporarily insolvent firms, Japanese banks continued extending credit to firms unlikely to repay the loans. The practice of keeping insolvent firms alive—even when receipt of repayment is doubtful—is referred to as evergreening. Evergreening was especially prevalent in the Japanese financial system. Bank loans actually increased in underperforming sectors (Hoshi and Kashyap 2004).

Firms with lower rates of profit and poor rates of return in the market received more loans on the whole. Evergreening continued despite the lack of profitability due to the overvaluation of *keiretsu* loans; had these firms gone bankrupt, the overvaluation of capital on bank balance sheets would be revealed (Hoshi and Kashyap 2004). Whenever borrowers faced serious financial trouble, banks would extend loans to conceal the status of borrowers' underperforming loans (Fukuda and Nakamura 2011).

The Japanese government also contributed to this process. After 1998, the government heavily encouraged banks to increase their lending to small and medium-sized firms, providing subsidized credit to these firms. Evergreening kept firms alive which distorted competition through subsidized credit to underperforming firms. By the early 2000s, roughly 30% of publicly traded firms were on "life support," though weighted by assets, these firms only constituted 15% of all publicly traded firms (Caballero, Hoshi, and Kashyap 2008). These firms—referred to as zombie firms—exhibited inefficient behavior, impacting productivity performance in Japan as inefficient firms reduced sector growth (Ahearne and Shinada 2005). These zombie firms created ongoing distortions in the market with lower job creation and industry productivity (Caballero, Hoshi, and Kashyap 2008).

Simultaneously, in an attempt to raise capital from the private sector, banks would raise money from life insurance companies, in a style of cyclical asset purchasing to build capital known as double gearing. Banks issued securities to be purchased by life insurance corporations. In return, life insurance companies issued subordinated debt and notes purchased by banks which counted towards capitalization ratios for banks while simultaneously increasing the capital with which life insurers could purchase bank issuances. With this transaction, banks and insurers appeared more capitalized on their balance sheets than the real amount of capitalization (Hoshi and Kashyap 2004). At the end of March 2001, seven life insurers held ¥5.4 trillion of bank stocks, and ¥5.1 trillion of bank subordinated debts. In exchange, banks held ¥1 trillion of insurers' surplus notes and ¥1.7 of subordinated debt (Fukao 2003).

These transactions introduced systemic risk to the Japanese banking sector, with insurance companies constituting at least two of many banks' top five shareholders as of 2002 (BIS 2002). Any failures of these insurers will lead to direct losses by banks (Hoshi and Kashyap 2004). When Chiyoda Life failed in October of 2000, Tokai Bank lost ¥74 billion yen—more than 10% of the size of Tokai Bank's capital injection in March 1999 (Fukao 2003; DICJ 2020b).

While Japanese regulators prohibit this transaction between pairs of banks or pairs of insurers, they allow the behavior between bank-insurer combinations (Hoshi and Kashyap 2004). However, during the early 2000s, this practice was actually encouraged by certain regulators, with the head of the Financial Services Agency publicly stating that double gearing is highly beneficial in increasing public confidence in financial institutions (Fukao 2003).

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